three at the rear of the vehicle detect any car or pedestrian 360 degrees around the car. The measurements are done by emitting laser pulses in the invisible infrared region. The echo of the light pulses is detected, and its flight time determines the distance to objects. A rotating laser scanner on top of the vehicle provides additional measurements – up to one million scan points per second of the 3D structure of the environment. A black and white video camera behind the rear-view mirror is

Professor Andrew Ng, (center) and his graduate students Pieter Abbeel (left) and Adam Coates have developed an artificial intelligence system that enables "autonomous" helicopters to teach themselves to fly by watching the maneuvers of a radio-control helicopter flown by a human pilot.

used to detect the white lane strips and centre the car on its lane. Two colour cameras are used to identify traffic lights and their state. During the test drive in Berlin, the car processed 46 traffic lights successfully during each of the four runs.

According to project leader professor Raul Rojas, autonomous vehicles could be already used in private roads since the technology is mature. Driving autonomously on public highways could become acceptable within the next ten to fifteen years, once thorny legal issues have been sorted out. Navigating autonomously in the streets requires additional technological development in order to make the cars extremely safe.